Unit 3: Measurement — Predicted Test Questions

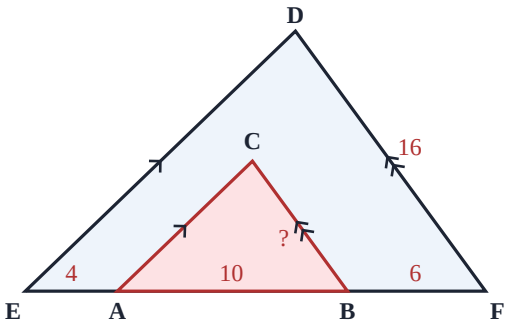
Math 9E · Lessons 3.1–3.4

These 28 questions are the ones your teacher is most likely to ask, based on your worksheets: the ★★★/★★★★ practice questions, the proofs done in class and the "Challenge before next class". Each question has a short note explaining why it is likely. Try them as a practice test first; worked answers are at the end.

Questions

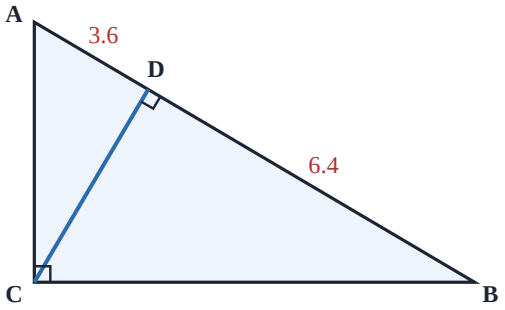
Lesson 3.1 — Similar Triangles

Q1 In ΔEFD, points A and B lie on EF with EA = 4, AB = 10 and BF = 6. Point C is inside the triangle with AC ∥ ED and BC ∥ FD. FD = 16. (a) Prove ΔABC ∼ ΔEFD. (b) Find BC.



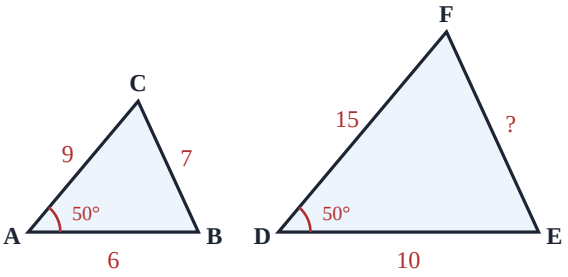
Why: Practice 3.1 #9 asks you to prove two triangles similar and then find x.

Q2 In right ΔABC (right angle at C), CD ⊥ AB. Prove $CD^2 = AD \cdot DB$. Then, if AD = 3.6 and DB = 6.4, find CD, AC and BC.



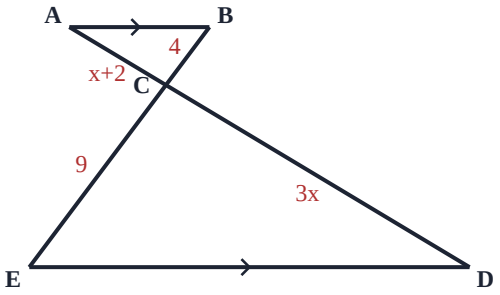
Why: you were asked in class to prove $CD^2 = AD \times DB$. Expect a proof followed by numbers.

Q3 In ΔABC, AB = 6, AC = 9 and ∠A = 50°. In ΔDEF, DE = 10, DF = 15 and ∠D = 50°. (a) Prove the triangles are similar. (b) If BC = 7, find EF.



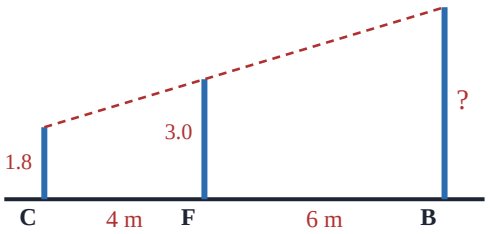
Why: SAS similarity is in your notes and is easy to test as a proof.

Q4 In an hourglass figure, AD and BE cross at C and AB ∥ ED. AC = x + 2, CD = 3x, BC = 4 and CE = 9. Find x, AC and CD.



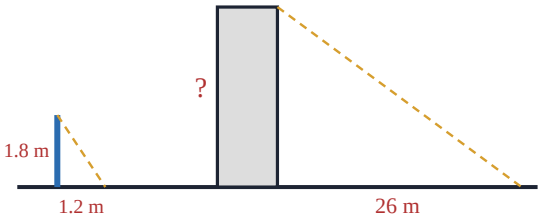
Why: Practice 3.1 #6 (hourglass). Adding algebra (x) is the usual way to make it harder.

Q5 Three vertical posts stand at points C, F and B on level ground, with CF = 4 m and FB = 6 m. The post at C is 1.8 m tall and the post at F is 3.0 m tall. The tops of all three posts lie on one straight line. How tall is the post at B? (Hint: subtract 1.8 m to make similar triangles.)



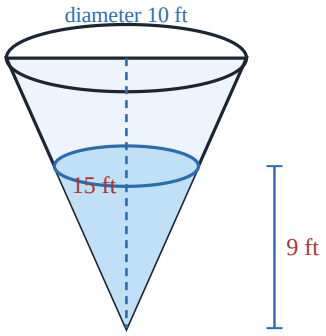
Why: Practice 3.1 #8 (three vertical posts).

Q6 A 1.8 m pole casts a 1.2 m shadow. At the same time a building casts a 26 m shadow. (a) Find the building's height. (b) Later, the building's shadow is 39 m. How long is the pole's shadow then?



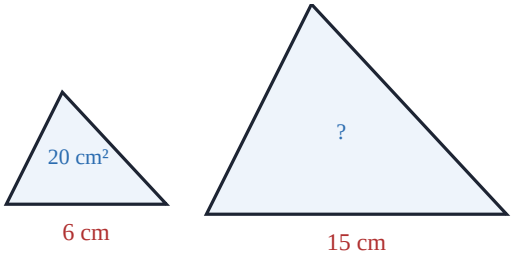
Why: the flagpole and Delbert problems (Example 3 and #4), with a second step.

Q7 A conical tank (point down) is 15 ft deep with top diameter 10 ft. It is filled to a depth of 9 ft. Find (a) the area of the water surface and (b) the volume of water.



Why: Practice 3.1 #5 (water surface in a conical tank), with volume added.

Q8 Corresponding sides of two similar triangles are 6 cm and 15 cm. The small triangle has area 20 cm². Find the area of the large one.



Why: Practice 3.1 #7 (★★★+) area ratio k^2, in reverse.

Q9 Two similar cones have volumes 54π cm³ and 250π cm³. (a) Find the ratio of their radii. (b) The small cone has surface area 36π cm². Find the surface area of the large cone.

Why: a natural step up from #7: from areas (k^2) to volumes (k^3).

Lesson 3.2 — Unit Conversions & Surface Area

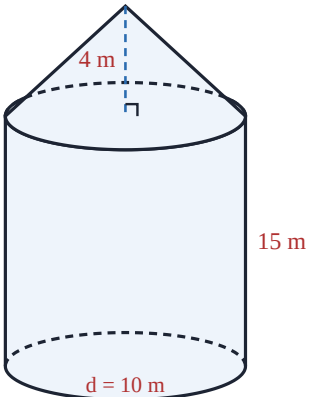
Q10 A doctor orders 0.6 g of syrup medicine. The label reads 250 mg in 5 mL. How many mL should be given? How many teaspoons is that (1 tsp = 5 mL)?

Why: Practice 3.2 #3–#4. The trap is the g → mg conversion.

Q11 1 ft = 30.48 cm. (a) How many cm² are in 1 ft²? (b) A room is 12 ft × 15 ft. Find its area in m², to the nearest hundredth.

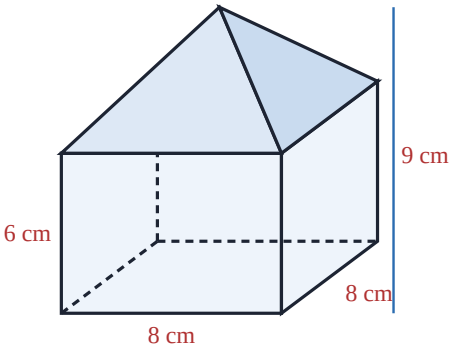
Why: Practice 3.2 #2g converts ft² to mm². Squaring the conversion factor is the main trap.

Q12 A silo is a cylinder with a conical roof. Diameter 10 m, cylinder height 15 m, roof height 4 m. The walls and roof need sealant (not the base). One bucket covers 30 m² and costs $65. Find the area (2 decimal places), number of buckets and total cost.



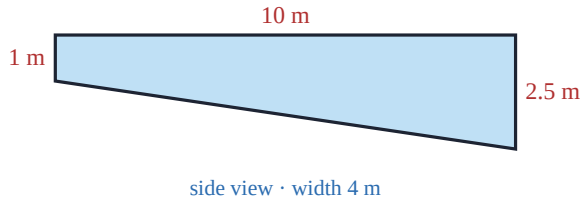
Why: Practice 3.2 #8 (silo sealant). You must find the slant height with Pythagoras first.

Q13 A square pyramid sits on top of a box with an 8 cm × 8 cm base and walls 6 cm tall. The total height of the shape is 9 cm. Find the total surface area.



Why: Practice 3.2 #11 (pyramid on a box) with new numbers.

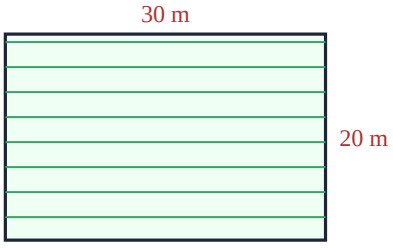
Q14 A pool is 10 m long and 4 m wide. The depth slopes from 1 m to 2.5 m. Tile the bottom, both side walls and both end walls. (a) Find the area to be tiled. (b) Tiles cost \$40 USD/m² and 1 USD = 1.38 CAD. Find the cost in CAD.



side view · width 4 m

Why: Practice 3.2 #10 (pool tiles) adds a currency conversion.

Q15 A rectangular lawn is 30 m × 20 m. A mower cuts a 50-cm strip and mows back and forth along the 30 m length. Find the distance travelled in km and the time in minutes at 4 km/h.

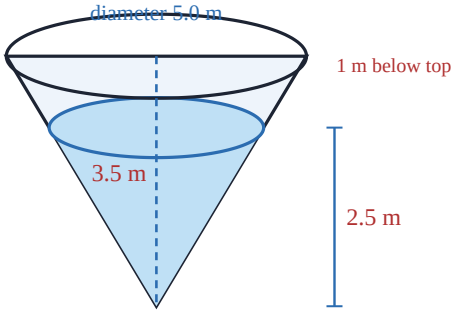


50 cm strips

Why: Practice 3.2 #9 (★★★★ lawn mower), the same reasoning on a simpler shape.

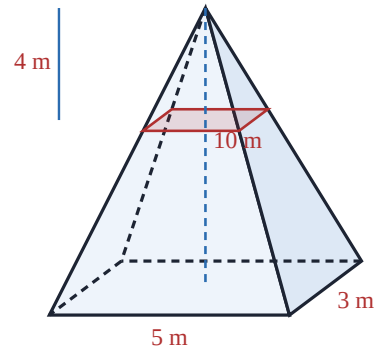
Lesson 3.3 — Volume & Capacity

Q16 An underground tank is a right cone with its apex pointing down. Base diameter 5.0 m, height 3.5 m. (a) Find its capacity in kL to the nearest tenth. (b) How much water is in it when the water level is 1 m below the top? (c) What percentage full is it?



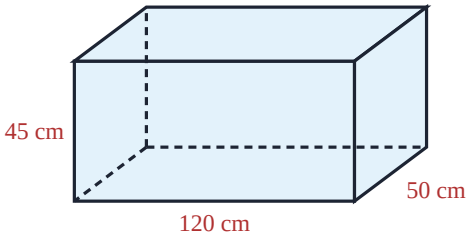
Why: Practice 3.3 #4b (★★★+), the "1 m below the top" twist that needs k³.

Q17 A right rectangular pyramid has base 5 m × 3 m and height 10 m. A horizontal cut is made 4 m from the apex and the top pyramid is removed. Find the volume of the remaining piece.



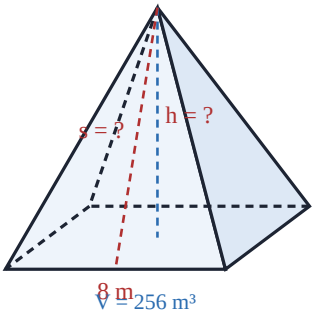
Why: Practice 3.3 #9 (★★★+), cut pyramid with different numbers.

Q18 A fish tank is 120 cm × 50 cm × 45 cm. Fancy goldfish need at least 75 US gallons (1 gal = 3.785 L). (a) Is the tank big enough? (b) Keeping length and width the same, what height would hold exactly 75 gallons?



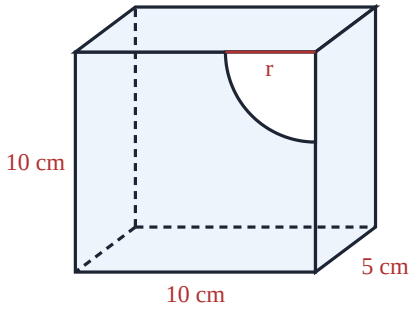
Why: Practice 3.3 #6 (fish tank in gallons) with a reverse step.

Q19 A right square pyramid has volume 256 m³ and base side 8 m. Find its height and its slant height to the nearest hundredth.



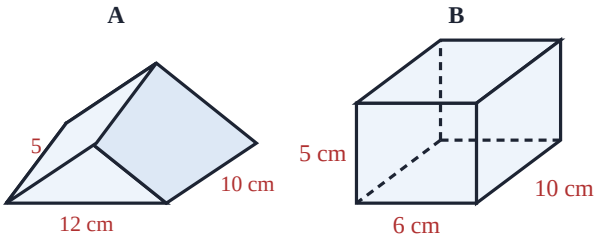
Why: Practice 3.3 #5. You find the height from the volume, then the slant height.

Q20 A quarter cylinder is cut from one edge of a 10 cm × 10 cm × 5 cm cuboid. The quarter cylinder runs the 5 cm length. The remaining volume is 400 cm³. Find the radius.



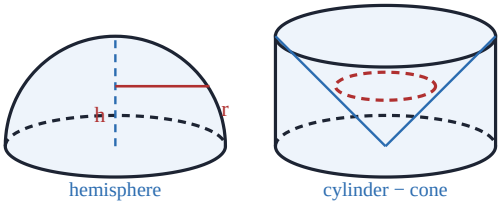
Why: Practice 3.3 #8 (quarter cylinder cut from a cuboid).

Q21 Prism A has a triangular cross-section (base 12 cm, height 5 cm) and is 10 cm long. Prism B is 6 cm × 5 cm × 10 cm. Carl says they cannot have the same volume because the shapes are different. Is he right? Use Cavalieri's principle.



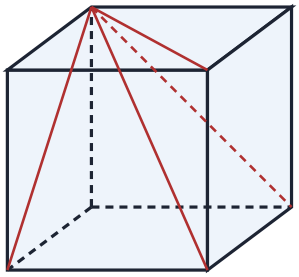
Why: Practice 3.3 #1 (Carl vs Hiva, Cavalieri).

Q22 **Derivation.** Use Cavalieri's principle with a hemisphere and a cylinder with a cone removed (same radius r and height r) to derive $V_{sphere} = (4/3)\pi r^3$.



Why: Practice 3.3 #13 (★★★★), deriving the sphere formula.

Q23 **Derivation.** Explain why the volume of a pyramid is one-third of the volume of a prism with the same base and height.

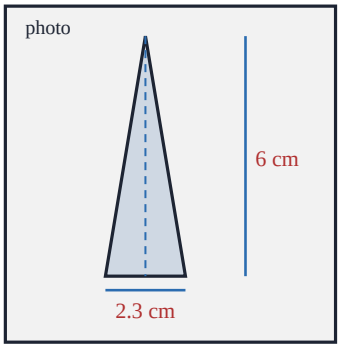


3 pyramids share one corner

Why: the "Challenge before next class" on your answer-key page.

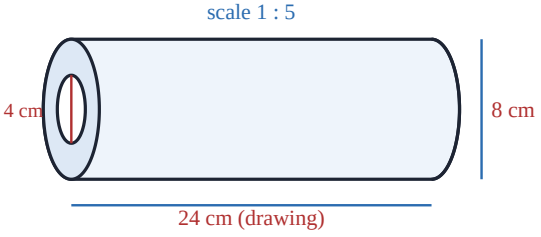
Lesson 3.4 — Scale & Scale Diagrams

Q24 A photo shows a tower. On the photo the tower is 6 cm tall and its base is 2.3 cm wide. The real tower is 324 m tall. (a) Write the scale of the photo as a ratio 1 : n. (b) Find the real width of the base.



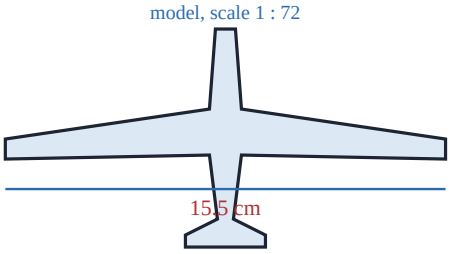
Why: Practice 3.4 #4 (Golden Gate photo) uses this exact two-step pattern.

Q25 A scale diagram of a pipe has scale 1 : 5. On the diagram the pipe is 24 cm long, its outer diameter is 8 cm and the circular opening is 4 cm across. Find (a) the real length, (b) the real diameter of the opening and (c) the real volume of material in the pipe.



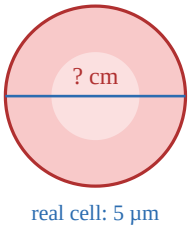
Why: Practice 3.4 #7 (★★★) asks exactly these three parts. Part (c) mixes scale with the volume of a cylinder.

Q26 A collectable model plane has scale 1 : 72. The model's wingspan is 15.5 cm. Find the real wingspan in the most appropriate SI unit.



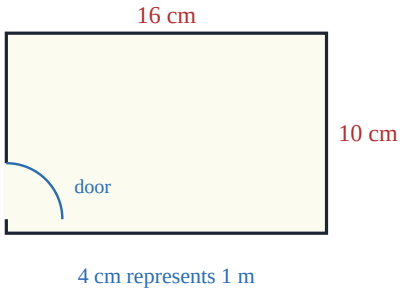
Why: Practice 3.4 #3 is the same question with a different measurement.

Q27 A red blood cell has diameter 5 µm. It is drawn at a scale of 8000 : 1. What diameter should the drawing have, in cm?



Why: Practice 3.4 #2 asks you to draw an 8000:1 diagram of a cell, which needs this calculation first.

Q28 A bathroom is drawn on grid paper where 4 cm represents 1 m. On the plan it is 16 cm × 10 cm. The walls are 3 m tall and have one door of 0.9 m × 2.1 m that is not painted. Tiling costs \$225/m² and painting costs \$16/m². Find the total cost including 12% tax.



Why: Practice 3.4 #8 (★★★★ bathroom): scale, area, perimeter and cost.

Answers & Worked Solutions

Lesson 3.1 — Similar Triangles

Q1 (a) AC ∥ ED ⇒ ∠CAB = ∠DEF (corresponding angles). BC ∥ FD ⇒ ∠CBA = ∠DFE. So ΔABC ~ ΔEFD (AA). (b) EF = 4 + 10 + 6 = 20, k = AB/EF = 10/20 = 1/2 ⇒ BC = ½ · FD = **8**.

Q2 Proof: ∠A = ∠DCB (both equal 90° − ∠ACD) and both triangles have a right angle at D ⇒ ΔADC ~ ΔCDB (AA) ⇒ AD/CD = CD/DB ⇒ CD^2 = AD · DB ∎. Numbers: CD^2 = 3.6 × 6.4 = 23.04 ⇒ **CD = 4.8**. AB = 10. AC^2 = 3.6 × 10 = 36 ⇒ **AC = 6**. BC^2 = 6.4 × 10 = 64 ⇒ **BC = 8**.

Q3 (a) DE/AB = 10/6 = 5/3 and DF/AC = 15/9 = 5/3, and the included angles are equal (∠A = ∠D = 50°) ⇒ **ΔABC ~ ΔDEF (SAS)**. (b) EF = 7 × 5/3 ≈ **11.67**.

Q4 ΔABC ~ ΔDEC ⇒ AC/DC = BC/EC: (x + 2)/(3x) = 4/9 ⇒ 9x + 18 = 12x ⇒ **x = 6**, **AC = 8**, **CD = 18**.

Q5 Above 1.8 m, the rise is 1.2 m over 4 m. Similar triangles: rise to B / 10 = 1.2/4 ⇒ 3 m. Post at B = 1.8 + 3 = **4.8 m**.

Q6 (a) H/26 = 1.8/1.2 ⇒ **H = 39 m**. (b) Now height = shadow (39 m : 39 m), so the pole's shadow = **1.8 m**.

Q7 r = 5 × 9/15 = 3 ft. (a) π(3^2) = **9π ≈ 28.27 ft^2**. (b) ⅓π(9)(9) = **27π ≈ 84.82 ft^3**.

Q8 k = 15/6 = 2.5 ⇒ k^2 = 6.25. Area = 20 × 6.25 = **125 cm^2**.

Q9 (a) k^3 = 250/54 = 125/27 ⇒ k = **5/3**, so radii are in the ratio 3 : 5. (b) Area ratio k^2 = 25/9 ⇒ 36π × 25/9 = **100π cm^2**.

Lesson 3.2 — Unit Conversions & Surface Area

Q10 0.6 g = 600 mg. 600 × 5/250 = **12 mL = 2.4 tsp**.

Q11 (a) 30.48^2 ≈ **929.03 cm^2** (b) 180 ft^2 × 0.3048^2 m^2 ≈ **16.72 m^2**.

Q12 r = 5. Roof slant = √(5^2 + 4^2) = √41 ≈ 6.403. Walls 2π(5)(15) = 150π; roof π(5)(6.403) ≈ 32.02π. Total ≈ **571.82 m^2**. 571.82 ÷ 30 = 19.06 ⇒ **20 buckets ⇒ $1300**.

Q13 Pyramid height = 9 − 6 = 3. Slant = √(3^2 + 4^2) = 5. Walls 4(8)(6) = 192; base 64; triangles 4 · ½ · 8 · 5 = 80. Total = **336 cm^2**.

Q14 (a) Sloped bottom: length √(10^2 + 1.5^2) ≈ 10.11 m × 4 ≈ 40.45 m^2. Two trapezoid sides: 2 × (1 + 2.5)/2 × 10 = 35 m^2. Ends: 1 × 4 + 2.5 × 4 = 14 m^2. Total ≈ **89.45 m^2**. (b) 89.45 × 40 × 1.38 ≈ **$4937.50 CAD**.

Q15 Rows = 20 ÷ 0.5 = 40. Distance = 40 × 30 = 1200 m = **1.2 km**. Time = 1.2 ÷ 4 = 0.3 h = **18 min**.

Lesson 3.3 — Volume & Capacity

Q16 (a) V = ⅓π(2.5^2)(3.5) ≈ 22.91 m^3 ≈ **22.9 kL**. (b) Water depth = 2.5 m ⇒ k = 2.5/3.5 = 5/7. V = 22.91 × $(5/7)^3$ ≈ **8.3 kL**. (c) $(5/7)^3$ ≈ **36.4%**. Being only 1 m below the top still leaves the tank well under half full!

Q17 V_{big} = ⅓(15)(10) = 50 m^3. k = 4/10 = 0.4 ⇒ V_{small} = 50 × 0.064 = 3.2 m^3. Remaining = **46.8 m^3**.

Q18 (a) 120 × 50 × 45 = 270 000 cm³ = 270 L ÷ 3.785 ≈ 71.3 gal. **No, it is not big enough.** (b) 75 × 3.785 = 283.875 L = 283 875 cm³. h = 283 875 ÷ (120 × 50) ≈ **47.3 cm**.

Q19 ⅓(64)h = 256 ⇒ **h = 12 m**. Slant = √(12² + 4²) = √160 ≈ **12.65 m**.

Q20 Cuboid = 500 ⇒ removed = 100 = ¼πr²(5) ⇒ r² = 80/π ≈ 25.46 ⇒ **r ≈ 5.05 cm**.

Q21 Triangle area = ½(12)(5) = 30 cm². Rectangle area = 6 × 5 = 30 cm². Same cross-sectional area along the same 10 cm length ⇒ both volumes = **300 cm³**. Carl is **wrong**: the area must match, not the shape.

Q22 Slice both shapes at height h. Hemisphere slice: radius √(r² − h²) (Pythagoras), so area = π(r² − h²). Washer: the outer radius is r. The cone (apex at the bottom, height r, radius r) has radius h at height h, so washer area = πr² − πh² = π(r² − h²). The areas are equal at every height, so by Cavalieri $V_{hemisphere} = V_{cyl} - V_{cone}$ = πr³ − ⅓πr³ = ⅔πr³. Sphere = 2 × ⅔πr³ = **(4/3)πr³** ∎

Q23 A cube of side s can be split into **3 identical pyramids** that share one corner as their apex. Each has a square base s × s and height s, so V = s³/3 = ⅓(base area)(height). Stretching the cube into a box (or using Cavalieri) keeps the ratio 1/3, so $V_{pyramid}$ = **⅓Bh**.

Lesson 3.4 — Scale & Scale Diagrams

Q24 (a) 6 cm : 32 400 cm ⇒ **1 : 5400**. (b) 2.3 × 5400 = 12 420 cm = **124.2 m**.

Q25 (a) 24 × 5 = **120 cm**. (b) 4 × 5 = **20 cm** (outer diameter 40 cm). (c) V = π(20² − 10²)(120) = 36 000π ≈ **113 097 cm³ ≈ 113.1 L**.

Q26 15.5 × 72 = 1116 cm = **11.16 m**.

Q27 5 µm × 8000 = 40 000 µm = 40 mm = **4 cm**.

Q28 Room 4 m × 2.5 m. Floor = 10 m² × \$225 = \$2250. Walls: perimeter 13 m × 3 m = 39 m² − 1.89 m² (door) = 37.11 m² × \$16 = \$593.76. Subtotal \$2843.76; with tax ×1.12 = **\$3185.01**.